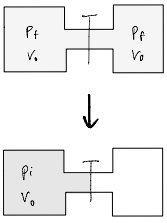
Maxwell's demon

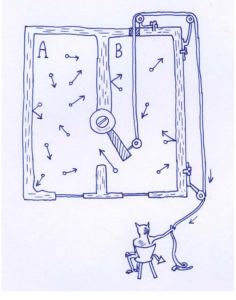
- How would we reverse a Joule expansion



- Push all the molecules on the right to the left → via isothermal compression

$\Delta T = 0$, $\Delta u = 0$, $dW = -\int p dV = -dQ$

$$\Delta W = \int_{2V_0}^{V_0} p\,dV = \int_{2V_0}^{V_0} \frac{R}{V} T\,dV = RT\ln 2 \Rightarrow T\Delta S_{gas}$$



- An intelligent being opens & closes a trap door connecting two chambers. Allowing only particles going from B → A, effectively reversing the Joule expansion without doing work. → in the pdV sense.

→ Seems as if a demon could cause entropy to decrease. making a mockery of 2nd Law.

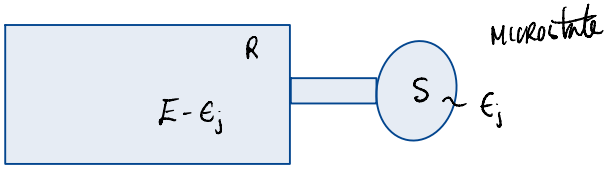
- Could be in the act of measurement / Act of opening the door } Devise schemes to do this with arbitrarily little work

- Now accepted answer is that the demon needs to have a memory of particles it has measured.

- After a while to make for more information the demon has to erase some of it. This erasure of information costs entropy.

- Information & Entropy → Next class

Boltzmann Revisited



$$P_U(E) = \frac{1}{\Omega(E)} = \frac{1}{\sum_{\epsilon_j} \Omega_R(E-\epsilon_j)\,\Omega(\epsilon_j)} = \frac{1}{\sum_{\epsilon_j} \Omega_R(E-\epsilon_j)} \propto 1.$$

$$P_j = \sum_{\substack{\text{overall possible} \\ \text{microstates of} \\ \text{reservoir that} \\ \text{have system} \\ \text{in } \epsilon_j}} P_U(E) = \underbrace{P_U(E)}_{\text{constant}} \Omega_R(E-\epsilon_j) \propto \Omega_R(E-\epsilon_j)$$

$$\frac{\Omega_R(E-\epsilon_j)}{\Omega(E)}$$

→ Now use $\left.\frac{d \ln \Omega_R(E)}{dE}\right|_{E=\bar{E}} = \frac{1}{k_B T}$

where we used $\frac{1}{k_B T} = \frac{d \ln \Omega(E)}{dE}$ which come from the assumption that system will maximize microstates available to it.

$\frac{d\Omega}{dE} = \frac{d}{dE}\left[\Omega(E)\Omega(\epsilon_j)\right] = 0$

$$\log\left(\Omega_R(E-\epsilon_j)\right) = \log\left(\Omega_R(\bar{E})\right) - \epsilon_j \frac{d \ln \Omega_R(E)}{dE}$$
$$\log\left(\Omega_R(E-\epsilon_j)\right) = \log\left(\Omega_R(E)\right) - \epsilon_j / k_B T$$

But

$$\log(P_j) = \log P_U(E) + \log\left(\Omega_R(E-\epsilon_j)\right)$$
$$= -\log \Omega(E) + \log\left(\Omega_R(E)\right) - \epsilon_j / k_B T$$
$$= \log\left(\frac{\Omega_R(E)}{\Omega(E)}\right) - \epsilon_j / k_B T$$

$$\Rightarrow \quad P_j = C\, e^{-\epsilon_j / k_B T}$$

$$C = \frac{\Omega_R(E)}{\Omega(E)} = \frac{\Omega_R(E)}{\sum_j \Omega(E-\epsilon_j)} \quad \text{which has to be} = \frac{1}{\sum_j e^{-\epsilon_j / k_B T}}$$

Entropy & probability

From B+B

* Recall Boltzmanns definition of entropy

$$S = k_B \ln \Omega$$

Entropy of a system in a particular macrostate in terms of Ω, the number of microstates associated with this macrostate. "Macrostate of fixed energy" - microcanonical ensemble

$\Omega(E)$

* Suppose a total system can have $N \{= \Omega(E)\}$ different equally likely microstates.

Hard to measure details

* These microstates are divided up into macrostates with $n_i [= \Omega(E-\epsilon_i)]$ microstates in the ith macrostates. Easy to measure

* We have $\sum_i n_i = N$ $\quad \{ \sum_j \Omega(E-\epsilon_j) = \Omega(E) \}$

* The probability P_i of finding the system in the ith macrostate is

$$P_i = n_i / N \qquad \text{So that} \qquad \sum P_i = 1$$

* The total entropy is $S_{Tot} = k_B \ln N$ but hard to figure out

* S_{Tot} = Entropy associated with freedom of being in different macrostates (S)

+ Entropy associated of being able to be in any one of different microstates within a macrostate $(S)_{micro}$

$$S_{tot} = S + S_{MICRO}$$

$$S_{MICRO} = \langle S_i \rangle = \sum_i P_i S_i = k_B \sum_i P_i \ln n_i$$

Where $S_i = k_B \ln n_i$ & P_i is the probability of a particular macrostate being occupied

$$S = S_{TOT} - S_{MICRO}$$
$$= k_B \left(\ln N - \sum P_i \ln n_i \right)$$
$$= k_B \sum_i P_i \left(\ln N - \ln n_i \right)$$

& using $\ln N - \ln n_i = \ln \left(\frac{N}{n_i} \right) = -\ln P_i$ & we get

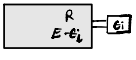
$$\boxed{S = -k_B \sum_i P_i \ln P_i}$$

(Gibbs expression)

Recap

B+B "P_i is probability to be in ith macrostate"

& Be careful → they mean the macrostate as defined as the many configurations of R such that ϵ_i is the exact arrangement of the system. Lets call that number $n_i = \Omega_R(E-\epsilon_i)$



$\sum n_i = N = \Omega(E)$ & $P_i = n_i / N = \frac{\Omega_R(E-\epsilon_i)}{\Omega(E)}$

total number of possible configurations is N.

$S_{TOTAL} = k \log N$

$S_{MICRO} = \langle S_i \rangle = -k \sum P_i \log n_i$ ← mean entropy associated with n_i configurations

$S = S_{TOTAL} - S_{MICRO} = k \left(\log N - \sum_i P_i \log n_i \right) = k \left(\sum P_i \left(\log N - \log n_i \right) \right) = -k \sum P_i \log \left(\frac{n_i}{N} \right) = -\sum P_i \log (P_i)$

If we take a system Ω states with prob $P_i = 1/\Omega$

Then ~ $P_i \propto e^{-E_i / k_B T}$

$$S = -k_B \sum_i P_i \ln P_i = -k_B \sum_{i=1}^{\Omega} \frac{1}{\Omega} \ln \frac{1}{\Omega} = k_B \ln \Omega$$

which is just like Boltzmann expression

due to David Tong.

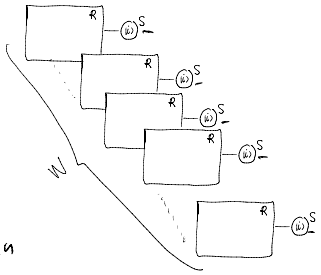
Another way

System in any state $|i\rangle$ with prob p_i. Ensemble of W systems. Number of states in $|i\rangle$'s is now $p_i W$.

turned probability into a number

p_i to be in a microstate $|i\rangle$

W ensemble



$\Omega_i = W p_i$ # of states in W with S being in the $|i\rangle^{th}$ microstate.

$\Omega = \dfrac{W!}{\prod_i W p_i!}$ # of arrangements of W such that we have

$W p_1$ in $|1\rangle$
$W p_2$ in $|2\rangle$
$W p_3$ in $|3\rangle$
$\vdots$
$W p_N$ in $|N\rangle$

W now microcanonical. How many ways to put $p_i W$ systems in $|i\rangle$ for each $|i\rangle$: $\Omega \simeq \dfrac{W!}{\prod_i p_i W!}$

$$
\begin{aligned}
S &= k_B \ln \Omega \\
&= k_B \left(\ln W! - \sum_i \ln (p_i W)! \right) \\
&= k_B \left(W \ln W - W - \sum_i \left[(p_i W) \ln (p_i W) - p_i W \right] \right) \\
&= k_B \left(W \ln W - \sum_i (p_i W) \ln (p_i W) \right) \\
&= k_B \left(W \ln W - \sum_i p_i W \left(\ln p_i + \ln W \right) \right) \\
&= k_B \left(W \ln W - \sum_i p_i W \ln W - \sum_i (p_i W) \ln p_i \right) \\
&= -k_B W \sum_i p_i \ln p_i
\end{aligned}
$$

N particles

$\binom{N}{n_1}\binom{N-n_1}{n_2}\binom{N-n_1-n_2}{n_3} \quad N = n_1 + n_2 + n_3$

$W = \dfrac{N!}{(N-n_1)!\,n_1!} \cdot \dfrac{(N-n_1)!}{(N-n_1-n_2)!\,n_2!} \cdot \dfrac{(N-n_1-n_2)!}{(N-n_1-n_2-n_3)!\,n_3!}$

$W = \dfrac{N!}{n_1! n_2! n_3!} \Rightarrow \dfrac{N!}{\prod_i n_i!} \qquad \sum n_i = N$

Entropy is of W systems: $= -k_B W \sum_i p_i \ln p_i \Rightarrow$ Entropy per system $= -k_B \sum_i p_i \ln p_i$

Jaynes – Maximum Entropy principle

* When making inferences based on incomplete information, we should use the probability distribution that has the maximum entropy consistent with available information

* A way to get Boltzmann factor from Gibbs def of entropy using Lagrange multipliers.

Example 14.7

Maximize $S = -k_B \sum_i P_i \ln P_i$ (eqn 14.48) subject to the constraints that $\sum P_i = 1$ and $\sum_i P_i E_i = U$.

Solution:

Use the method of Lagrange multipliers,[6] in which we maximize

$$\frac{S}{k_B} - \alpha \times (\text{constraint 1}) - \beta \times (\text{constraint 2}) \tag{14.50}$$

where α and β are Lagrange multipliers. Thus we vary this expression with respect to one of the probabilities P_j and get

$$\frac{\partial}{\partial P_j}\left(\sum_i -P_i \ln P_i - \alpha P_i - \beta P_i E_i\right) = 0, \tag{14.51}$$

so that

$$-\ln P_j - 1 - \alpha - \beta E_j = 0. \tag{14.52}$$

This can be rearranged to give

$$P_j = \frac{e^{-\beta E_j}}{e^{1+\alpha}}, \tag{14.53}$$

so that with $Z = e^{1+\alpha}$ we have

$$P_j = \frac{e^{-\beta E_j}}{Z}, \tag{14.54}$$

which is our familiar expression for the Boltzmann probability (eqn 4.13).

$\beta \rightarrow$ constraint on average energy

$\alpha \rightarrow$ constraint on normalisation

} makes sense

Information Theory

	Prob of being true
(1) Isaac Newton's birthday falls on a particular day of the year.	1
(2) Isaac Newton's birthday falls in the second half of the year.	1/2
(3) Isaac Newton's birthday falls on the 25th of a month.	12/365

as P goes down Q (information content goes up)

* 2 & 3 are independent → so information content should add.

* Prob of (2) & (3) both being true, in absence of any prior info, is $P(2) \times P(3) = \frac{6}{365}$

* So, Probability of 2 independent statements being true is a product & natural to assume information content is additive, then we can understand why Shannon (1916–2001) adopted the following

The information content Q of a statement is defined by

$$Q = -k \log P$$

where P is probability of a statement being true & $k > 0$.

(if we use $\log \to \log_2$ & $k = 1$ then Q is in bits. If we $k = k_B$ & $\log \to \log_e = \ln$, then we get one that matches our entropy.)

* So if we have a set of statements with probability P_i with corresponding information content $Q_i = -k \log P_i$ then average information content is

$$S = \langle Q_i \rangle = \sum_i P_i Q_i = -k \sum_i P_i \log P_i$$

This is called the Shannon Entropy.

* How much information we gain, on average, following a measurement.
* Amount of uncertainty we have about a quantity before we measure.

Example 15.1

- A fair die produces outcomes 1, 2, 3, 4, 5, and 6 with probabilities $\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$. The information associated with each outcome is $Q = -k \log \frac{1}{6} = k \log 6$ and the average information content is then $S = k \log 6$. Taking $k = 1$ and using log to the base 2 gives a Shannon entropy of 2.58 bits.
- A biased die produces outcomes 1, 2, 3, 4, 5, and 6 with probabilities $\frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{2}$. The information contents associated with the outcomes are $k \log 10$, $k \log 10$, $k \log 10$, $k \log 10$, $k \log 10$, and $k \log 2$. (These are 3.32, 3.32, 3.32, 3.32, 3.32, and 1 bit respectively.) If we take $k = 1$ again, the Shannon entropy is then $S = k(5 \times \frac{1}{10} \log 10 + \frac{1}{2} \log 2) = k(\log \sqrt{20})$ (this is 2.16 bits). This Shannon entropy is smaller than in the case of the fair die.

Example 15.2

What is the Shannon entropy for a Bernoulli trial (a two-outcome random variable[5]) with probabilities P and $1 - P$ of the two outcomes?

Solution:

$$S = -\sum_i P_i \log P_i = -P \log P - (1 - P) \log(1 - P), \qquad (15.3)$$

where we have set $k = 1$. This behaviour is sketched in Fig. 15.1. The Shannon entropy has a maximum when $p = \frac{1}{2}$ (greatest uncertainty about the outcome, or greatest information gained, 1 bit, following a trial) and a minimum when $p = 0$ or 1 (least uncertainty about the outcome, or least information gained, 0 bit, following a trial).

The information associated with each of the two possible outcomes is also shown in Fig. 15.1 as dotted lines. The information associated with the outcome having probability P is given by $Q_1 = -\log_2 P$ and decreases as P increases. Clearly when this outcome is very unlikely (P small) the information associated with getting that outcome is very large (Q_1 is many bits of information). However, such an outcome doesn't happen very often so it doesn't contribute much to the average information (i.e., to the Shannon entropy, the solid line in Fig. 15.1). When this outcome is almost certain (P almost 1) it contributes a lot to the average information but has very little information content. For the other outcome, with probability $1 - P$, $Q_2 = -\log_2(1 - P)$ and the behaviour is simply a mirror image of this. The maximum average information is when $P = 1 - P = \frac{1}{2}$ and both outcomes have 1 bit of information associated with them.

[5] See Section 3.7

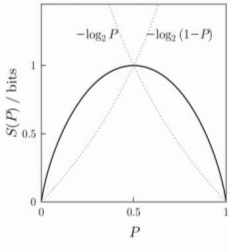


Fig. 15.1 The Shannon entropy of a Bernoulli trial (a two-outcome random variable) with probabilities of the two outcomes given by P and $1 - P$. The units are chosen so that the Shannon entropy is in bits. Also shown is the information associated with each outcome (dotted lines).